Equity Liquidity Absorption and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Liquidity Absorption (ELA), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by **?**. A value-weighted long/short trading strategy based on ELA achieves an annualized gross (net) Sharpe ratio of 0.50 (0.44), and monthly average abnormal gross (net) return relative to the **?** five-factor model plus a momentum factor of 17 (16) bps/month with a t-statistic of 2.34 (2.26), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, Change in equity to assets, Asset growth, change in net operating assets, Total accruals, change in ppe and inv/assets) is 16 bps/month with a t-statistic of 2.41.

1 Introduction

The following automatically generated report tests the asset pricing implications of Equity Liquidity Absorption (ELA), and its robustness in predicting returns in the cross-section of equities. It is produced using the methodology of **?**, from input data consisting of firm-month observations for the proposed predictor.[1]

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on U.S.-domiciled firms trading on major exchanges and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) following standard practice in the literature.signal construction relies on two key COMPUSTAT variables. Equity in earnings of unconsolidated subsidiaries (CEQL) represents the firm's proportionate share of earnings or losses from investments in unconsolidated subsidiaries, joint ventures, and other equity method investments where the firm exercises significant influence but not control. Cash and short-term investments (CHE) captures the firm's liquid assets, including cash, cash equivalents, and marketable securities that can be readily converted to cash. These variables provide insight into a firm's investment activities and liquidity position.construct the Equity Liquidity Absorption signal by calculating the year-over-year change in equity in earnings (CEQL), scaled by lagged cash and short-term investments. Specifically, we compute (CEQL(t) - CEQL(t-1)) / CHE(t-1), where t denotes the fiscal year-end. This ratio captures the extent to which changes in equity method investment income absorb or generate liquidity relative to the firm's available liquid resources.

[1]It used version v0.4.1 of the publicly available code repository at https://github.com/velikov-mihail/AssayingAnomalies. See more details at http://AssayingAnomalies.com.

A positive value indicates increasing equity earnings relative to cash holdings, potentially signaling growing profitability from strategic investments or increased capital deployment to affiliates. We calculate this measure using annual observations from fiscal year-end financial statements and require non-missing values for both CEQL and CHE in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ELA signal. Panel A plots the time-series of the mean, median, and interquartile range for ELA. On average, the cross-sectional mean (median) ELA is -4.96 (-0.35) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input ELA data. The signal's interquartile range spans -2.56 to 0.68. Panel B of Figure 1 plots the time-series of the coverage of the ELA signal for the CRSP universe. On average, the ELA signal is available for 6.33% of CRSP names, which on average make up 7.74% of total market capitalization.

4 Does ELA predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ELA using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ELA portfolio and sells the low ELA portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the **?** three-factor model (FF3) and its variation that adds momentum (FF4), the **?** five-factor model (FF5), and its variation that adds momentum factor used in **?** (FF6). The table shows that the long/short ELA strategy earns an average return of 0.32% per month with a t-statistic of 3.93. The annualized Sharpe ratio of the strategy is 0.50. The alphas range from 0.17% to 0.33% per month and have t-statistics exceeding 2.34 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the **?** six-factor model. The long/short strategy's most significant loading is 0.65, with a t-statistic of 13.09 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 586 stocks and an average market capitalization of at least $1,296 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 28 bps/month with a t-statistics of 3.77. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of **?**. These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from **?**. The net average returns reported in the first column range between 23-40bps/month. The lowest return, (23 bps/month), is achieved from the quintile sort using name breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.75. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ELA trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-one cases.

Table 3 provides direct tests for the role size plays in the ELA strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ELA, as well as average returns and alphas for long/short trading ELA strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the ELA strategy achieves an average return of 37 bps/month with a t-statistic of 4.20. Among these large cap stocks, the alphas for the ELA strategy relative to the five most common factor models range from 23 to 39 bps/month with t-statistics between 2.77 and 4.41.

5 How does ELA perform relative to the zoo?

Figure 2 puts the performance of ELA in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[2] The vertical red line shows where the Sharpe ratio for the ELA strategy falls in the distribution. The ELA strategy's gross (net) Sharpe ratio of 0.50 (0.44) is greater than 90% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ELA strategy (red line).[3] Ignoring trading costs, a \$1 invested in the ELA strategy would have yielded \$7.57 which ranks the ELA strategy in the top 3% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ELA strategy would have yielded \$5.37 which ranks the ELA strategy in the top 2% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and **?** net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ELA relative to those. Panel A shows that the ELA strategy gross alphas fall between the 58 and 64 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ELA strategy has a positive net generalized alpha for five out of the five factor models. In these cases ELA ranks between the 75 and 83 percentiles in terms of how much it could have expanded the

[2]The anomalies come from March, 2022 release of the **?** open source asset pricing dataset.

[3]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in **?**, that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

achievable investment frontier.

6 Does ELA add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ELA with 210 filtered anomaly signals.[4] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ELA or at least to weaken the power ELA has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ELA conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ELA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ELA} ELA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ELA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ELA. Stocks are finally grouped into

[4]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

five ELA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ELA trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ELA and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ELA signal in these Fama-MacBeth regressions exceed 1.99, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on ELA is 2.22.

Similarly, Table 5 reports results from spanning tests that regress returns to the ELA strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ELA strategy earns alphas that range from 15-18bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.01, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ELA trading strategy achieves an alpha of 16bps/month with a t-statistic of 2.41.

7 Does ELA add relative to the whole zoo?

Finally, we can ask how much adding ELA to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following **?**. The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the ELA signal.[5] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes ELA grows to $2940.85.

8 Conclusion

This study provides compelling evidence that Equity Liquidity Absorption (ELA) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous testing protocol of Novy-Marx and Velikov (2023), demonstrates that ELA-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on ELA delivers an annualized Sharpe ratio of 0.50 gross (0.44 net of transaction costs), indicating strong economic significance that survives practical implementation considerations. Most notably, the strategy generates a statistically significant alpha of 17 basis points per month (t-statistic = 2.34) relative to the Fama-French five-factor model augmented with momentum, declining only marginally to 16 basis points after transaction costs. The robustness of these results is further confirmed by the persistence of abnormal returns (16 bps/month, t-statistic = 2.41) even after controlling

[5]We filter the 207 **?** anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ELA is available.

for six closely related strategies from the factor zoo, including various measures of asset growth, accruals, and investment patterns.

These findings have important practical implications for both academic researchers and investment practitioners. The economic magnitude and statistical significance of the ELA signal suggest it captures a distinct dimension of expected returns not fully explained by existing factor models or related anomalies. For practitioners, the strategy's resilience to transaction costs and its orthogonality to established factors indicate potential diversification benefits and implementation viability in real-world portfolios.

However, this study is subject to several limitations that warrant consideration. First, while we follow best practices in accounting for transaction costs, actual implementation costs may vary across different market conditions and institutional settings. Second, the sample period and geographic scope of our analysis may limit the generalizability of findings to other markets or time periods. Third, while we control for numerous related factors, the underlying economic mechanism driving the ELA effect requires further theoretical development.

Future research should explore several promising avenues. First, investigating the economic channels through which equity liquidity absorption affects returns could provide deeper insights into the pricing mechanism. Second, examining the international evidence and time-series stability of the ELA effect would enhance our understanding of its pervasiveness and robustness. Third, analyzing the interaction between ELA and market conditions, such as periods of financial stress or varying liquidity regimes, could reveal important conditional patterns. Finally, exploring the optimal combination of ELA with other factors in portfolio construction could yield valuable insights for practical implementation. Understanding whether the ELA premium compensates for systematic risk or represents a market inefficiency remains an important question for future investigation.

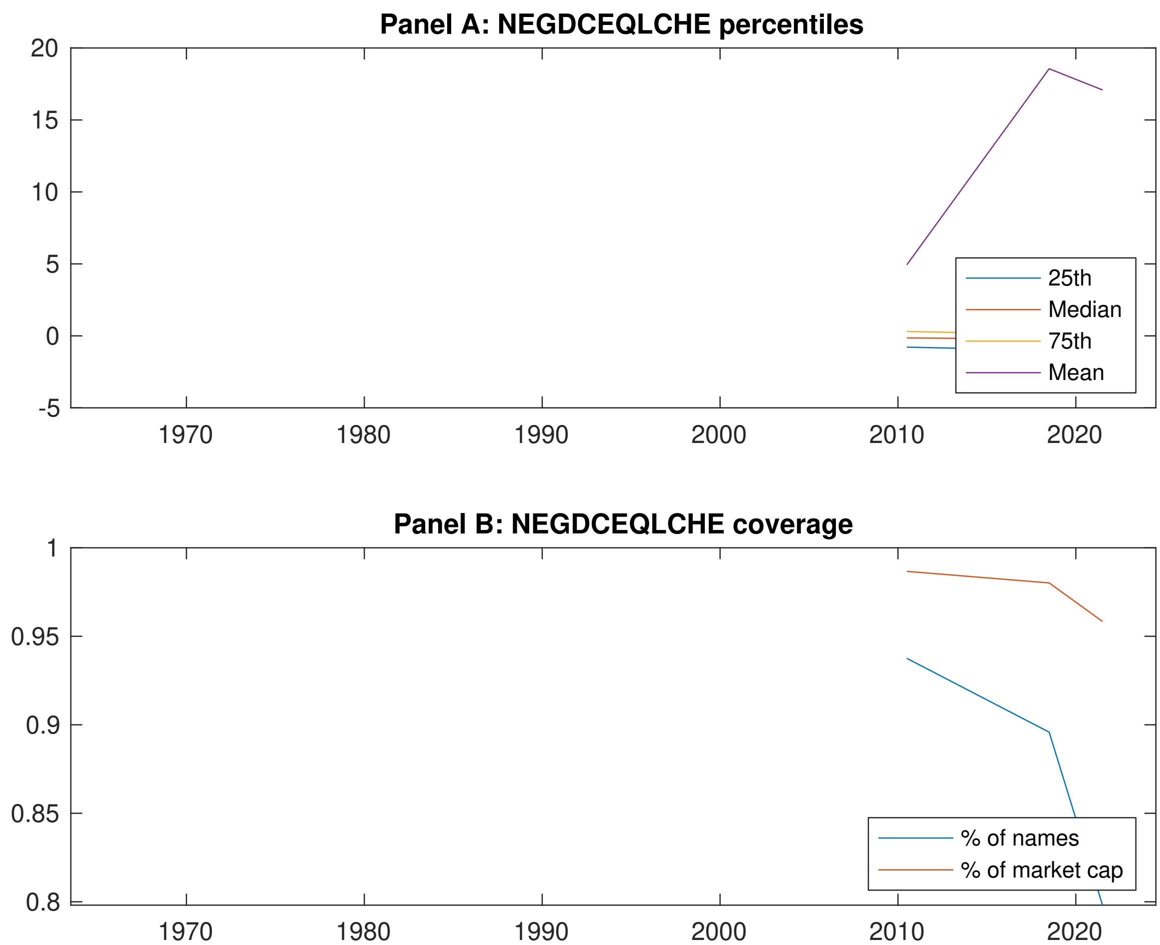


Figure 1: Times series of ELA percentiles and coverage.
This figure plots descriptive statistics for ELA. Panel A shows cross-sectional percentiles of ELA over the sample. Panel B plots the monthly coverage of ELA relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ELA. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, **?** three-factor model, **?** three-factor model augmented with the **?** momentum factor, **?** five-factor model, and the **?** five-factor model augmented with the **?** momentum factor following **?**. Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the **?** five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on ELA-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.37	0.58	0.70	0.65	0.69	0.32
	[2.17]	[3.45]	[4.13]	[3.78]	[4.03]	[3.93]
α_{CAPM}	-0.21	-0.00	0.11	0.06	0.12	0.33
	[-3.92]	[-0.00]	[2.73]	[1.29]	[1.95]	[4.06]
α_{FF3}	-0.22	0.03	0.16	0.00	0.04	0.26
	[-4.11]	[0.84]	[3.90]	[0.03]	[0.62]	[3.24]
α_{FF4}	-0.21	0.04	0.15	0.01	0.02	0.23
	[-3.77]	[1.00]	[3.56]	[0.28]	[0.40]	[2.85]
α_{FF5}	-0.25	0.05	0.17	-0.04	-0.07	0.18
	[-4.66]	[1.18]	[4.17]	[-0.84]	[-1.28]	[2.50]
α_{FF6}	-0.23	0.05	0.16	-0.02	-0.06	0.17
	[-4.37]	[1.23]	[3.84]	[-0.52]	[-1.20]	[2.34]
Panel B: **?** 6-factor model loadings for ELA-sorted portfolios						
β_{MKT}	0.98	0.97	0.99	1.06	1.03	0.05
	[76.25]	[98.87]	[99.49]	[97.88]	[80.95]	[2.91]
β_{SMB}	0.06	-0.04	-0.09	-0.06	0.12	0.06
	[3.51]	[-3.13]	[-6.60]	[-3.97]	[6.71]	[2.31]
β_{HML}	0.08	-0.02	-0.05	0.13	-0.02	-0.10
	[3.21]	[-0.80]	[-2.55]	[6.00]	[-0.65]	[-2.83]
β_{RMW}	0.16	0.05	-0.01	0.05	0.03	-0.13
	[6.42]	[2.43]	[-0.36]	[2.20]	[1.27]	[-3.79]
β_{CMA}	-0.15	-0.16	-0.08	0.13	0.50	0.65
	[-4.08]	[-5.59]	[-2.82]	[4.18]	[13.86]	[13.09]
β_{UMD}	-0.02	-0.00	0.02	-0.02	-0.01	0.01
	[-1.35]	[-0.44]	[1.66]	[-1.87]	[-0.41]	[0.69]
Panel C: Average number of firms (n) and market capitalization (me)						
n	608	586	660	766	782	
me ($\$10^6$)	1369	2106	2769	2586	1296	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the ELA strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, **?** three-factor model, **?** three-factor model augmented with the **?** momentum factor, **?** five-factor model, and the **?** five-factor model augmented with the **?** momentum factor following **?**. Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and **?** generalized alphas as prescribed by **?**. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.32	0.33	0.26	0.23	0.18	0.17
			[3.93]	[4.06]	[3.24]	[2.85]	[2.50]	[2.34]
Quintile	NYSE	EW	0.46	0.46	0.39	0.41	0.48	0.50
			[4.49]	[4.42]	[3.98]	[4.16]	[5.55]	[5.75]
Quintile	Name	VW	0.28	0.29	0.20	0.17	0.11	0.10
			[3.26]	[3.44]	[2.48]	[2.08]	[1.51]	[1.37]
Quintile	Cap	VW	0.28	0.30	0.24	0.24	0.17	0.18
			[3.77]	[4.03]	[3.37]	[3.20]	[2.63]	[2.71]
Decile	NYSE	VW	0.45	0.48	0.38	0.34	0.25	0.23
			[4.31]	[4.53]	[3.72]	[3.26]	[2.63]	[2.43]
Panel B: Net Returns and **?** generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.28	0.29	0.22	0.21	0.17	0.16
			[3.42]	[3.51]	[2.78]	[2.57]	[2.34]	[2.26]
Quintile	NYSE	EW	0.25	0.25	0.18	0.20	0.24	0.26
			[2.33]	[2.27]	[1.75]	[1.93]	[2.69]	[2.88]
Quintile	Name	VW	0.23	0.26	0.18	0.16	0.11	0.11
			[2.75]	[3.04]	[2.20]	[1.99]	[1.58]	[1.50]
Quintile	Cap	VW	0.24	0.27	0.22	0.22	0.17	0.17
			[3.29]	[3.62]	[3.02]	[2.96]	[2.62]	[2.68]
Decile	NYSE	VW	0.40	0.43	0.34	0.31	0.25	0.24
			[3.81]	[4.04]	[3.32]	[3.07]	[2.62]	[2.51]

Table 3: Conditional sort on size and ELA

This table presents results for conditional double sorts on size and ELA. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ELA. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ELA and short stocks with low ELA .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

	ELA Quintiles					ELA Strategies					
Size quintiles	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.53	0.82	0.94	0.97	0.71	0.19	0.13	0.04	0.01	0.09	0.07
	[2.15]	[3.51]	[4.28]	[3.71]	[2.40]	[1.32]	[0.91]	[0.31]	[0.08]	[0.71]	[0.55]
(2)	0.66	0.73	0.86	0.96	0.89	0.23	0.25	0.18	0.18	0.28	0.28
	[2.79]	[3.19]	[3.91]	[4.50]	[3.72]	[2.24]	[2.41]	[1.79]	[1.74]	[3.05]	[2.98]
(3)	0.62	0.77	0.80	0.89	0.83	0.21	0.22	0.15	0.12	0.24	0.21
	[2.87]	[3.64]	[3.90]	[4.45]	[3.87]	[2.00]	[2.05]	[1.45]	[1.15]	[2.47]	[2.19]
(4)	0.51	0.67	0.79	0.87	0.83	0.32	0.29	0.19	0.23	0.14	0.18
	[2.63]	[3.43]	[4.01]	[4.52]	[4.13]	[3.67]	[3.33]	[2.33]	[2.69]	[1.70]	[2.17]
(5)	0.33	0.58	0.62	0.60	0.70	0.37	0.39	0.34	0.32	0.23	0.24
	[1.97]	[3.56]	[3.70]	[3.38]	[4.39]	[4.20]	[4.41]	[3.83]	[3.60]	[2.77]	[2.78]

Panel B: Portfolio average number of firms and market capitalization

	ELA Quintiles: Average n					ELA Quintiles: Average market capitalization ($\$10^6$)				
Size quintiles	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
(1)	378	379	379	373	358	36	38	34	29	23
(2)	106	106	106	106	106	56	56	56	56	55
(3)	77	77	77	77	76	98	97	99	98	97
(4)	65	65	65	64	64	209	208	213	209	210
(5)	60	60	60	60	60	1179	1684	1920	1845	1522

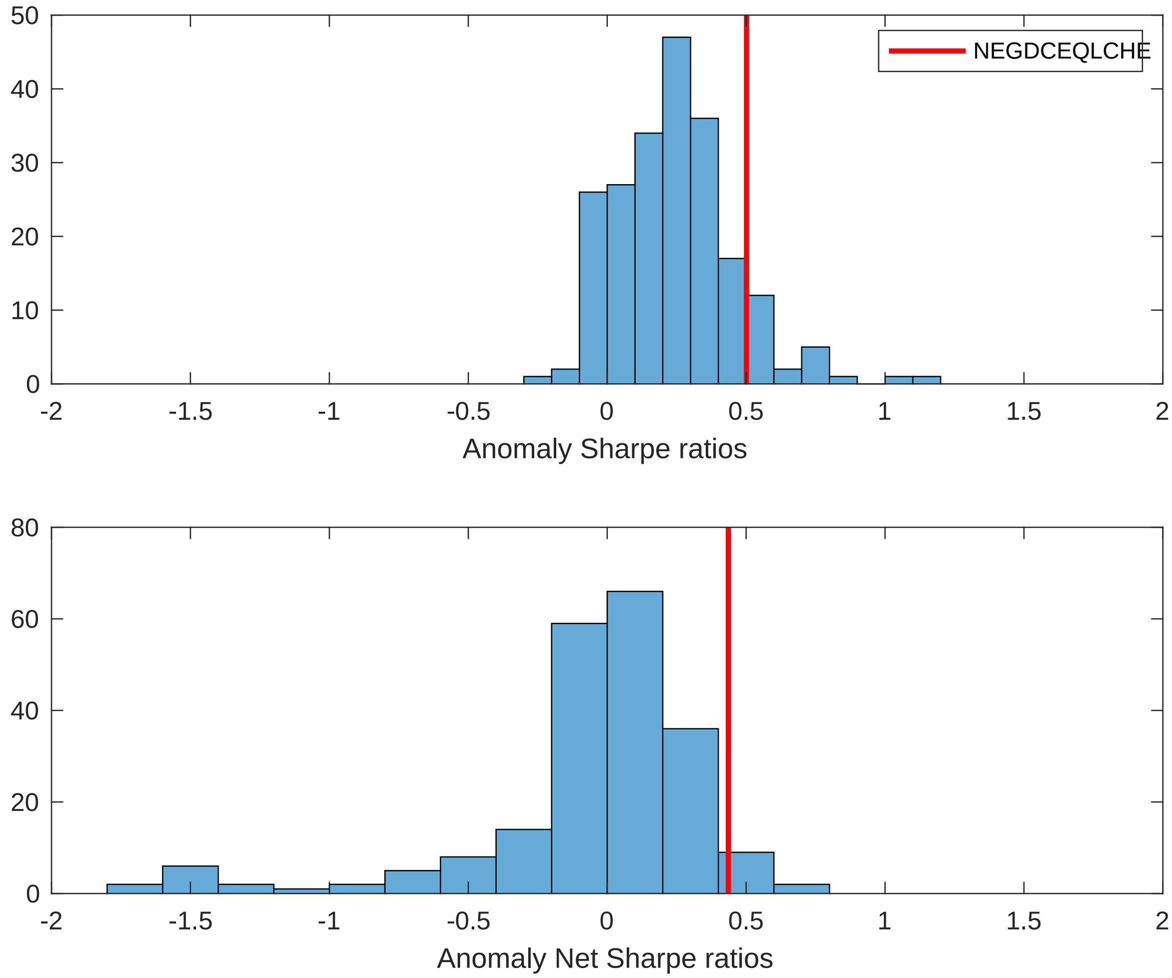


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ELA with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

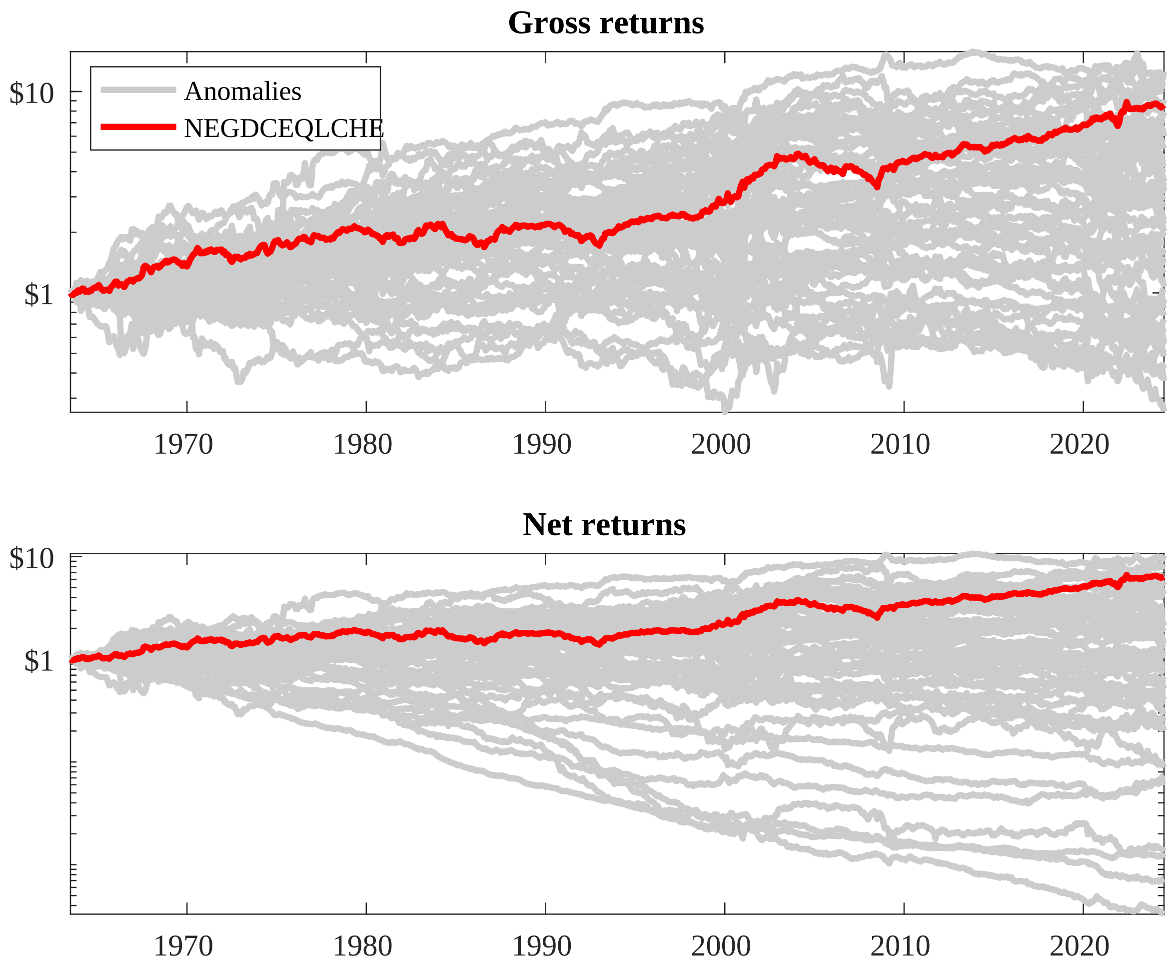
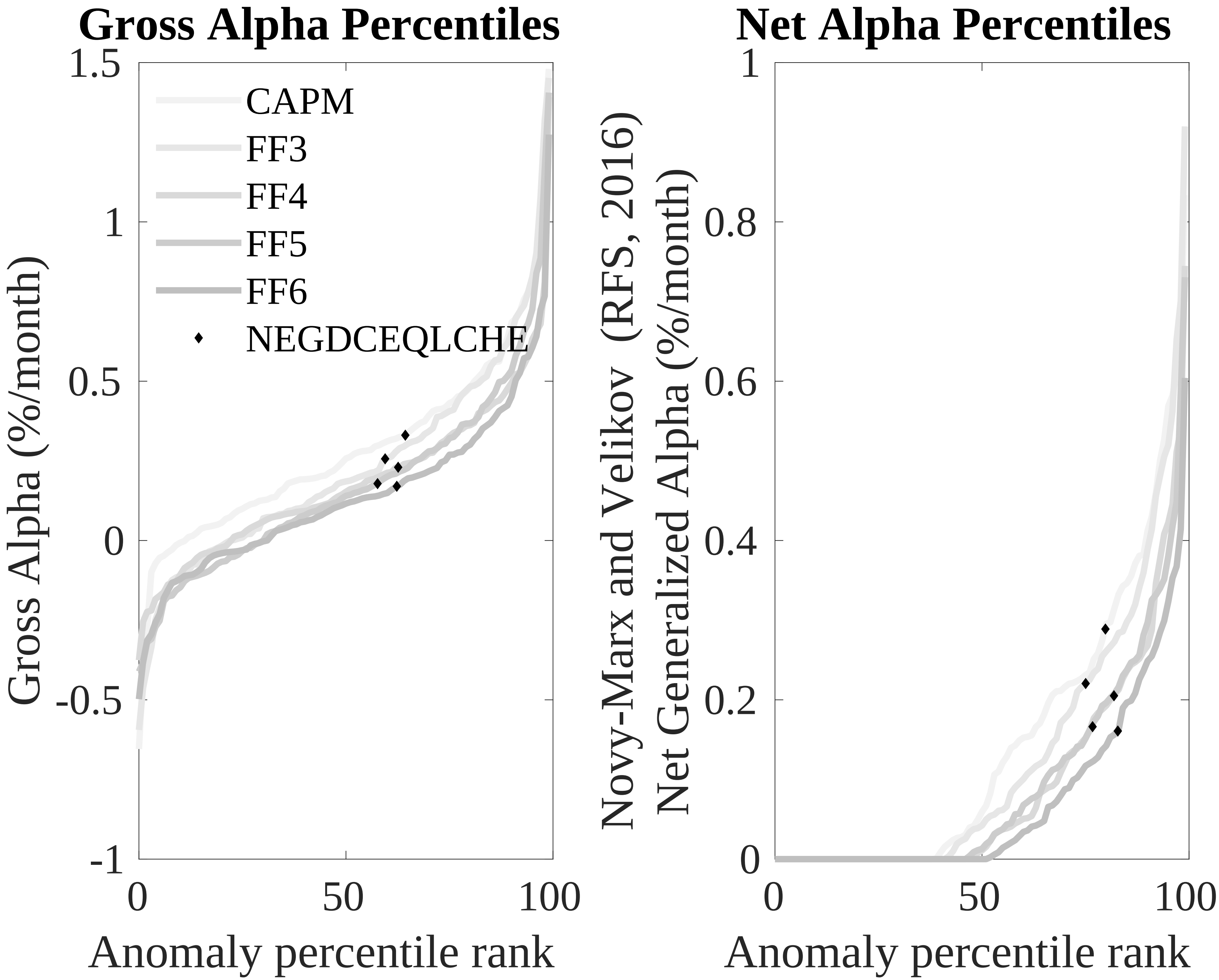
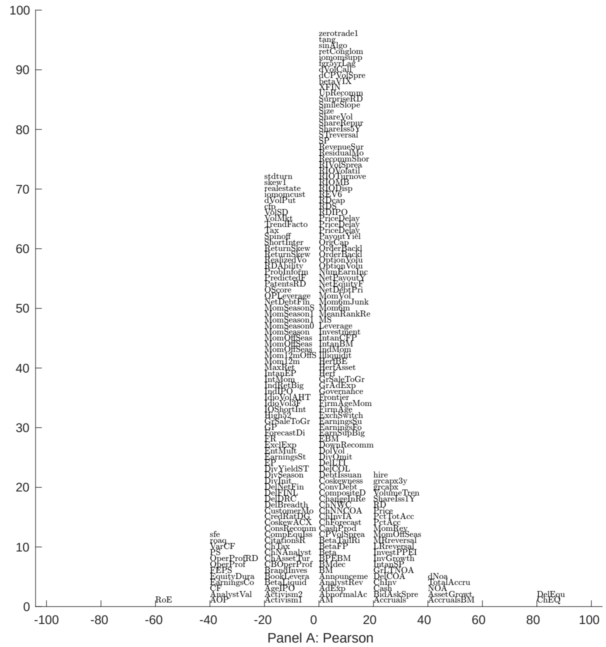


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ELA trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.



Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDCEQLCHE
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100



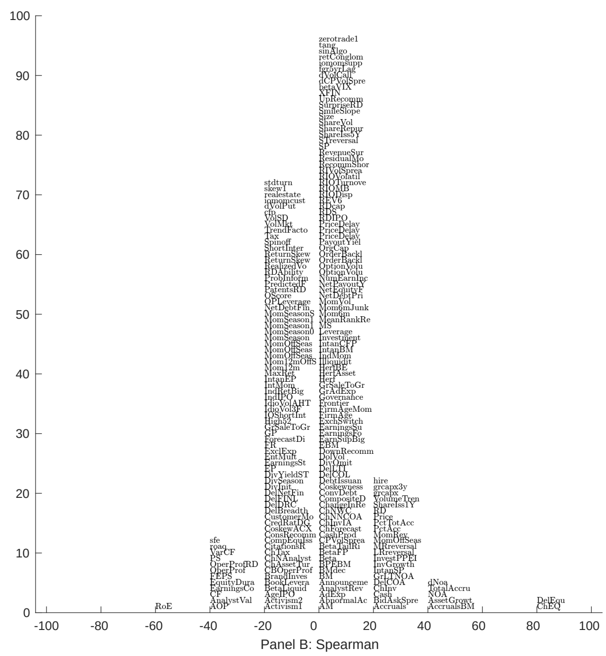


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with ELA. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

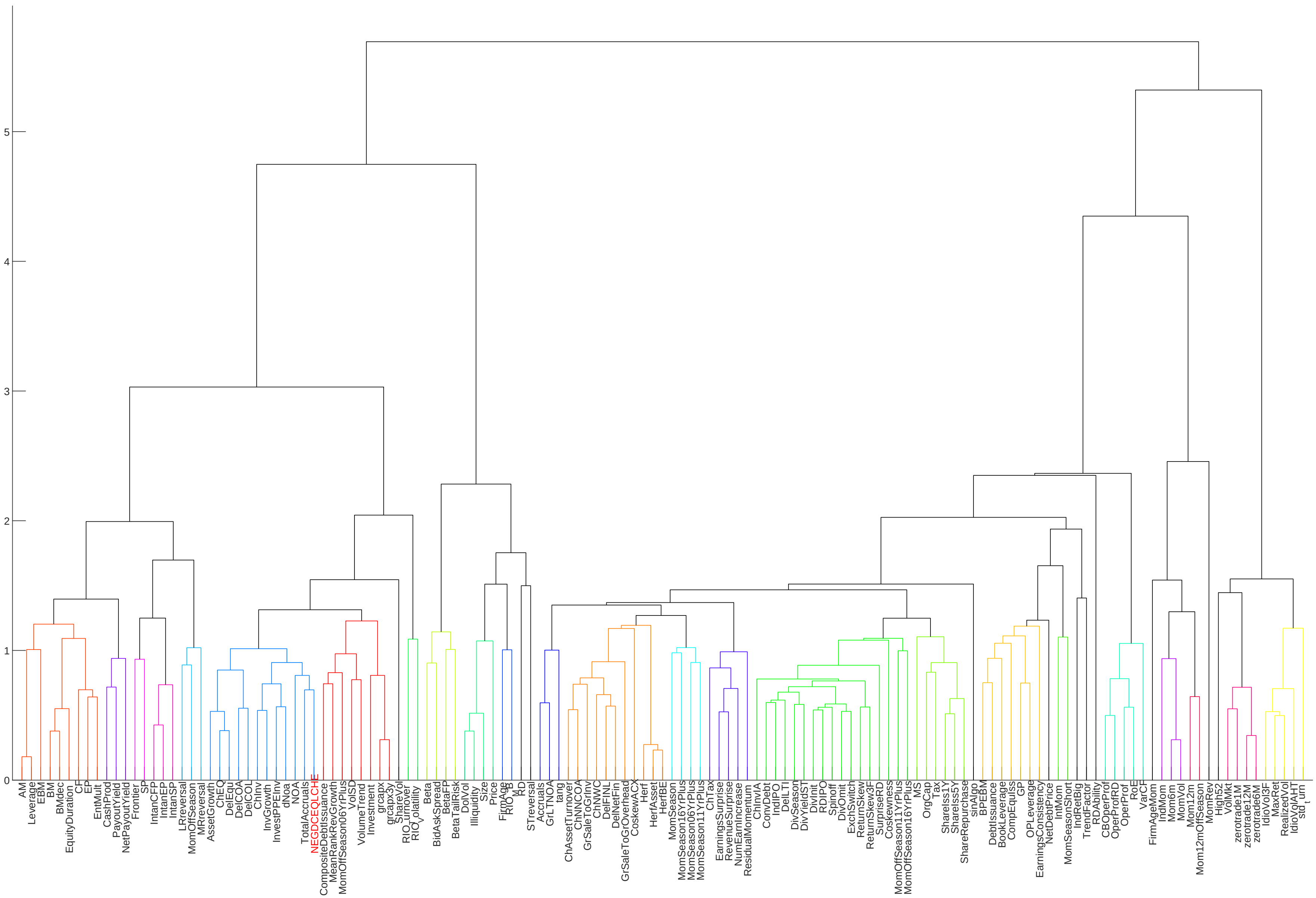


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

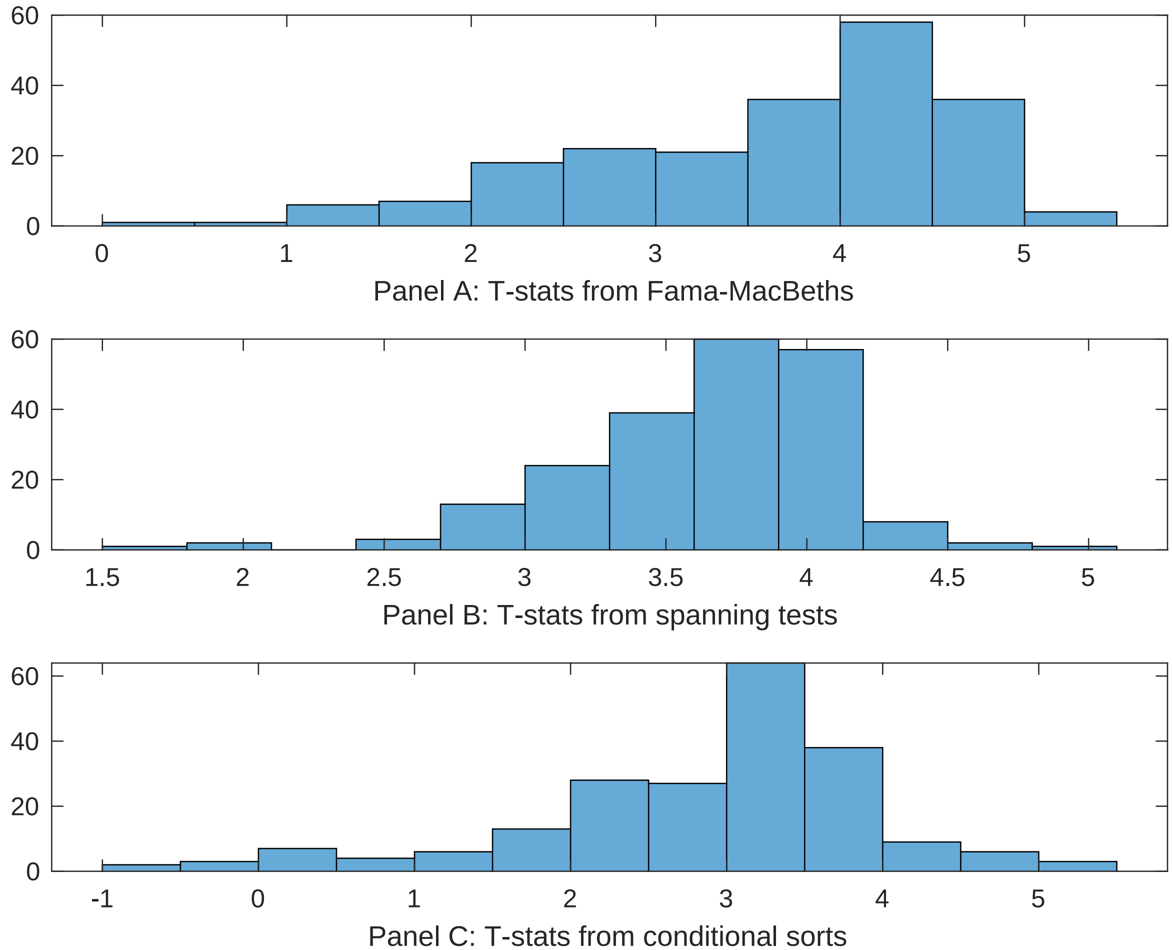


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ELA conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ELA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ELA} ELA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ELA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ELA. Stocks are finally grouped into five ELA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ELA trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on ELA. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ELA} ELA_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Growth in book equity, Change in equity to assets, Asset growth, change in net operating assets, Total accruals, change in ppe and inv/assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.17	0.13	0.13	0.13	0.12	0.14	0.12
	[7.26]	[5.73]	[6.29]	[6.03]	[5.58]	[6.13]	[5.85]
ELA	0.16	0.15	0.13	0.12	0.26	0.18	0.12
	[2.92]	[2.48]	[2.10]	[1.99]	[3.75]	[2.84]	[2.22]
Anomaly 1	0.41						-0.14
	[3.85]						[-0.95]
Anomaly 2		0.13					0.59
		[3.77]					[1.01]
Anomaly 3			1.00				0.32
			[9.59]				[1.63]
Anomaly 4				0.13			0.55
				[10.32]			[2.48]
Anomaly 5					0.41		-0.19
					[2.20]		[-0.78]
Anomaly 6						0.16	0.54
						[8.70]	[2.39]
# months	720	720	732	720	732	732	720
$\bar{R}^2$(%)	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ELA trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ELA} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the **?** five-factor model augmented with the **?** momentum factor. The six most closely related anomalies, X, are Growth in book equity, Change in equity to assets, Asset growth, change in net operating assets, Total accruals, change in ppe and inv/assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.15	0.18	0.17	0.15	0.15	0.17	0.16
	[2.26]	[2.63]	[2.43]	[2.01]	[2.11]	[2.37]	[2.41]
Anomaly 1	42.95						28.75
	[11.83]						[5.53]
Anomaly 2		37.32					21.29
		[11.11]					[4.12]
Anomaly 3			21.37				-6.18
			[4.99]				[-1.32]
Anomaly 4				16.88			6.53
				[4.15]			[1.50]
Anomaly 5					13.25		-5.67
					[3.98]		[-1.57]
Anomaly 6						12.70	5.34
						[3.75]	[1.60]
mkt	6.57	4.48	5.19	5.25	4.71	5.01	5.87
	[4.10]	[2.77]	[3.02]	[3.04]	[2.72]	[2.90]	[3.67]
smb	4.89	5.85	4.61	6.51	6.38	5.82	5.42
	[2.11]	[2.51]	[1.84]	[2.61]	[2.56]	[2.33]	[2.34]
hml	-14.23	-13.94	-9.65	-10.25	-8.81	-10.29	-16.28
	[-4.56]	[-4.42]	[-2.91]	[-3.06]	[-2.64]	[-3.06]	[-5.22]
rmw	-11.00	-9.62	-13.35	-12.90	-10.95	-13.11	-10.51
	[-3.52]	[-3.04]	[-3.99]	[-3.83]	[-3.21]	[-3.89]	[-3.36]
cma	22.56	26.60	38.38	51.58	57.53	54.44	16.22
	[3.93]	[4.67]	[5.40]	[8.95]	[11.10]	[9.79]	[2.35]
umd	0.47	2.12	1.92	0.91	1.58	1.25	0.75
	[0.30]	[1.32]	[1.12]	[0.53]	[0.92]	[0.73]	[0.47]
# months	732	732	732	732	732	732	732
$\bar{R}^2$(%)	40	38	30	30	30	29	41

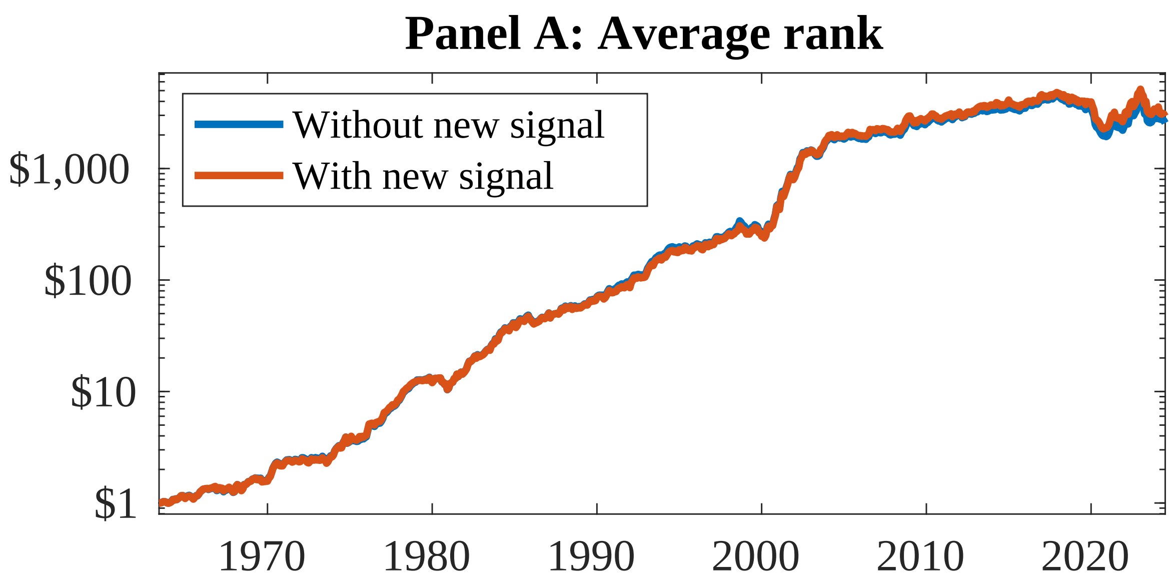


Figure 8: Combination strategy performance
This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following **?**. In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as ELA. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.